Department of Computer Science and Engineering

1. Subject Code: **AIS-601** Course Title: **Computational Intelligence - II**

2. **Expected Effort:** Total: 150 Hours Pacing: 10 Hours / Week Duration: 15 Weeks

3. **Prerequisites**
   - AIS-401

4. **Course Structure**
   5 Units × 30 Hours (lectures, source-code reading, laboratory, PoW)

1. Course Description

AIS-601 completes the AI track by moving from consuming models to adapting, scaling and governing them. Students fine-tune and align open-weight models, build multimodal and real-time voice pipelines, execute distributed training and optimised inference, engineer durable multi-agent platforms with sandboxed code execution, and finally impose security, evaluation and cost governance on the result. Every capability is delivered as an operable service with measured latency, quality and cost, and defended under an adversarial review.

2. Course Outcomes

After successful completion of the course, students will be able to:

- **CO1.** Fine-tune and align open-weight models using parameter-efficient methods and preference optimisation.
- **CO2.** Construct multimodal and real-time speech pipelines under explicit latency and quality budgets.
- **CO3.** Engineer distributed training and high-performance inference across multiple accelerators.
- **CO4.** Design agentic platforms with durable state, long-term memory and sandboxed tool execution.
- **CO5.** Operate AI systems securely and economically through red-teaming, evaluation gating, governance and cost engineering.

3. Detailed Syllabus

Unit	Contents	Hrs
1	**Model Adaptation: Datasets, PEFT & Preference Alignment** *Module A: Data Curation for Adaptation* Task framing and dataset design, instruction formatting and chat templates, tokenisation pitfalls, synthetic data generation and filtering, deduplication, contamination checks, and train/validation splitting for small datasets. *Module B: Parameter-Efficient Fine-Tuning* Full fine-tuning versus PEFT. LoRA, QLoRA and DoRA: rank, alpha, target modules and adapter merging. Supervised fine-tuning loops, learning-rate schedules, packing, and catastrophic forgetting mitigation. *Module C: Preference Optimisation and Evaluation* RLHF overview and its cost model. Direct Preference Optimisation and related objectives (ORPO, KTO). Reward hacking and over-optimisation. Benchmarking with standard evaluation harnesses and statistically honest reporting of deltas. **Hands-on Laboratory & Proof-of-Work (PoW)** • **PoW:** QLoRA fine-tune an 8B-class open-weight model on a curated domain dataset, then apply preference optimisation on collected preference pairs. • **PoW:** Benchmark base versus adapted model on a held-out suite and report effect sizes with confidence intervals, not single-run scores. • Merge adapters into a base checkpoint and verify serving parity.	30

Unit	Contents (continued)	Hrs
2	**Multimodal Systems: Vision, Speech & Generative Media** *Module A: Vision–Language Systems* Contrastive embedding models (CLIP/SigLIP) and multimodal retrieval. Vision–language model architecture: vision encoder, projector and language decoder. Document AI: layout parsing, OCR, table and figure extraction. *Module B: Speech Pipelines* Automatic speech recognition, streaming transcription and voice activity detection. Text-to-speech synthesis, barge-in handling, and end-to-end latency budgeting for conversational systems. *Module C: Generative Media* Diffusion fundamentals: forward/reverse process, samplers, classifier-free guidance. U-Net versus diffusion transformer backbones. Conditioning with ControlNet and image LoRAs. Provenance, watermarking and content-authenticity considerations. **Hands-on Laboratory & Proof-of-Work (PoW)** • **PoW:** Build a multimodal document question-answering service combining OCR, a vision–language model and hybrid retrieval, evaluated on a labelled document set. • **PoW:** Build a real-time voice agent (VAD → ASR → LLM → TTS) and demonstrate sub-second perceived response latency under load. • Fine-tune an image LoRA and evaluate fidelity and prompt adherence against the base model.	30
3	**Distributed Training & High-Performance Inference** *Module A: Parallelism Strategies* Data, tensor, pipeline and sequence parallelism. ZeRO stages and FSDP sharding. Gradient accumulation and checkpointing at scale. Collective communication primitives and interconnect bandwidth limits. *Module B: Training Reliability and Throughput* Checkpointing, resumption and fault tolerance. Throughput accounting (MFU), stragglers, dataloader bottlenecks, and mixed-precision numerics at scale. Experiment tracking and reproducibility. *Module C: Inference Optimisation* KV-cache internals and paged memory, prefix and prompt caching, speculative and multi-token decoding, structured/constrained decoding, multi-LoRA serving, batching policy tuning, and GPU autoscaling with queue-aware routing. **Hands-on Laboratory & Proof-of-Work (PoW)** • **PoW:** Train a model across multiple GPUs using FSDP and report scaling efficiency and model FLOPs utilisation against the single-device baseline. • **PoW:** Deploy a multi-LoRA inference server with speculative decoding and publish a latency/throughput/cost frontier for at least three configurations. • Instrument and eliminate a dataloader or communication bottleneck, evidencing the change with traces.	30

Unit	Contents (continued)	Hrs
4	**Agentic Platforms: Memory, Planning & Sandboxed Execution** *Module A: Agent Architectures* ReAct, Plan-and-Execute, and reflection-based architectures. Multi-agent orchestration, role specialisation, handoff protocols, and termination/budget conditions. Failure taxonomies and recovery. *Module B: Memory Systems* Short-term context management and compaction. Long-term episodic and semantic memory over vector plus graph stores. Retrieval-triggered memory, forgetting policies, and user-scoped isolation. *Module C: Safe Execution and Durability* Tool schema design and MCP server operation at scale. Executing untrusted generated code in sandboxes (seccomp, gVisor, Firecracker MicroVMs), egress control and resource caps. Durable workflow execution, checkpointing, retries and compensation. Human-in-the-loop approval gates. **Hands-on Laboratory & Proof-of-Work (PoW)** • **PoW:** Build a durable multi-agent platform that executes untrusted model-generated code inside a Firecracker MicroVM with checkpointed workflow state and full replay. • **PoW:** Implement a long-term memory subsystem with tenant isolation and demonstrate measurable task-success improvement over a stateless baseline. • Inject tool failures and network partitions; demonstrate correct recovery and no duplicated side effects.	30
5	**AI Security, Evaluation, Governance & FinOps** *Module A: Threat Modelling and Defence* Direct and indirect prompt injection, data exfiltration through tools, jailbreak taxonomies, insecure output handling, model and dataset supply-chain risk. Guardrail architectures, input/output classifiers, PII detection and redaction, and least-privilege tool design. *Module B: Evaluation and Assurance* Rubric and LLM-as-judge evaluation, golden datasets, regression suites gated in CI, canary evaluation in production, drift and hallucination monitoring, and incident triage for model-caused failures. *Module C: Governance and Cost Engineering* Model cards, data lineage and consent, retention policy, and an overview of the EU AI Act and ISO/IEC 42001 obligations. Cost modelling per token, per request and per GPU-hour; routing and cascading; batching and caching economics; showback reporting. **Hands-on Laboratory & Proof-of-Work (PoW)** • **PoW:** Red-team a deployed agent and publish a documented vulnerability report with reproductions, severity ratings and shipped mitigations. • **PoW:** Ship a CI-gated evaluation suite plus a live cost dashboard, and demonstrate a measured cost reduction with no quality regression. • Produce a model card and data-lineage record for the capstone system.	30

Unit	Contents (continued)	Hrs
	Total Instructional Effort	**150**

4. Recommended Resources

- Hu et al., *LoRA*; Dettmers et al., *QLoRA*; Rafailov et al., *Direct Preference Optimization*.
- Rajbhandari et al., *ZeRO*; official PyTorch FSDP and DeepSpeed documentation.
- Official vLLM, Hugging Face TRL/PEFT, Whisper and Firecracker documentation.
- OWASP Top 10 for LLM Applications; NIST AI Risk Management Framework.
- EU AI Act summary materials and ISO/IEC 42001 overview.